Instability reveals regulatory signals masked by pooled differential expression in heterogeneous conditions

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Abstract

Pooled differential expression analysis assumes coherent responses across samples, an assumption often violated in heterogeneous biological systems. When internal responses differ across subgroups or time, opposing or structured effects can collapse toward near-zero pooled estimates, obscuring biologically relevant regulation. Here, we introduce instability as a quantitative measure of internal disagreement across structured partitions. Using synthetic benchmarks, instability increases selectively under opposing and temporally structured regimes while remaining low in coherent and null conditions. In real stress-response datasets, instability identifies genes with strong subgroup-dependent or temporally varying responses that are attenuated or underrepresented in pooled analysis and are not sufficiently explained by variance alone. Across conditions, instability is elevated in the presence of structured heterogeneity and remains minimal in homogeneous data, indicating that it reflects internal disagreement rather than expression magnitude or dispersion. This leads to systematic divergence in gene prioritization, with instability-based ranking identifying gene sets largely distinct from those selected by pooled effect magnitude. Together, these results establish instability as a distinct analytical dimension for revealing structured regulatory signals masked by conventional pooling.

Keywords: Differential expression, heterogeneous transcriptomics, instability, structured heterogeneity, pooled analysis bias, subgroup conflict, temporal dynamics

1. Introduction

In large-scale and meta-analytic RNA-seq studies, heterogeneity is widespread because samples are generated in different technical contexts (library preparation, sequencing, and processing batches) and often represent diverse biological states, producing batch effects and variable responses that can rival biological signals of interest [1] . Quantifying differential gene expression under heterogeneous biological conditions remains an unresolved challenge despite the maturity of transcriptomic analysis pipelines. Standard workflows summarize gene-level responses into a single effect size by pooling all samples, implicitly assuming that the underlying biological response is coherent. This assumption underlies widely used frameworks such as DESeq2 and limma[1, 2], which estimate condition-level effects from aggregated data, as well as established RNA-seq analysis and benchmarking practices[3, 4].

This assumption breaks down when responses within a condition are not internally coherent. In the presence of subgroup-specific or time-dependent regulation, genes may exhibit effects that differ in both magnitude and direction across samples (Fig. 1a). Under these conditions, pooling no longer acts as a noise-reduction step but instead removes structured variation (Fig. 1b). As illustrated in Fig. 1, opposing or divergent internal responses can collapse toward a pooled estimate near zero, creating the appearance of weak or absent regulation despite strong underlying activity (Fig. 1c). This represents a systematic failure mode in which structured signals are suppressed by averaging, consistent with observations in heterogeneous biological systems and mixed cell populations[5-8]. Because pooled models provide no direct indication that response coherence has been violated, such failures can remain hidden in standard differential-expression workflows.

Existing approaches to heterogeneity extend conventional models through subgroup analysis, interaction terms, or time-course analysis, but do not directly quantify whether effects are consistent across groups[9-12]. While effective when structure is known, these methods depend on correct specification of subgroups or temporal relationships and do not quantify whether responses are mutually consistent across partitions. In many settings, such a structure is only partially observed or remains latent, particularly in complex or dynamic systems such as single-cell or pseudobulk analyses[13-16].

Variance-based measures do not resolve this limitation. Although variance captures dispersion, it does not distinguish between stochastic variability and structured divergence, and therefore cannot differentiate noise from conflicting subgroup responses[17-20]. As a result, current frameworks lack a measure of the degree to which internal responses agree or disagree prior to aggregation.

To address this limitation, we introduce instability as a quantitative measure of structured heterogeneity, capturing both effect divergence and directional inconsistency across partitions. In contrast to conventional differential expression frameworks, which assume a single consistent effect per gene, instability explicitly evaluates whether this assumption holds in heterogeneous data. Rather than compressing responses into a single pooled estimate, it preserves the relational structure among components of the response.

Treating instability as a primary analytical dimension changes the interpretation of heterogeneous data. Genes that appear weak under pooled analysis may exhibit high instability, reflecting structured regulatory divergence rather than the absence of signal, whereas genes with strong pooled effects but low instability represent coherent regulation.

In this study, heterogeneity refers to the internal structure within a condition rather than differences between treatment and control groups. We define instability as a quantitative measure of internal disagreement across structured partitions and use it to identify regulatory signals that are systematically masked under pooled differential expression. Using controlled synthetic benchmarks and real stress-response datasets, we show that instability captures a distinct class of heterogeneous responses that are not explained by variance alone and correspond to biologically interpretable forms of internal disagreement in heterogeneous biological systems[21-24].

2. Results

To evaluate instability across distinct forms of internal heterogeneity, we analyzed RNA-seq datasets representing subgroup- and time-dependent stress responses in *Ramazzottius varieornatus*. Ultraviolet exposure (UV) contained temporally distinct response regimes with partially conflicting effects, desiccation (DES) exhibited subgroup-dependent response magnitude differences, and gamma radiation (GAM) represented temporal divergence across stress progression. In contrast, low-temperature controls (LT) showed comparatively consistent responses and were used as a relatively homogeneous reference condition. Here, heterogeneity refers to structured internal variation within a condition, such as temporal phases or biological subgroups, rather than differences between treatment and control groups.

2.1. Pooled analysis attenuates structured signals across synthetic regulatory regimes

We evaluated pooled analysis under controlled synthetic regimes (Fig. 2a–c). Pooled estimates preserved signal in coherent scenarios (pooled: 1.101 ± 0.036; structured: 1.102 ± 0.036), but strongly attenuated structured signals. In opposing scenarios, the mean absolute effect decreased from 1.497 ± 0.019 (structured) to 0.283 ± 0.019 (pooled), an ~5.3-fold reduction. In the phase-structured time-course data, the pooled magnitude (0.251 ± 0.022) was reduced relative to the structured estimates (0.850 ± 0.012), an ~3.4-fold decrease, while the coherent signals remained stable (pooled: 0.999 ± 0.031; structured: 1.001 ± 0.031). Null scenarios showed low effects under both analyses (subgroup: 0.282 ± 0.008 vs 0.325 ± 0.007; time-course: 0.252 ± 0.006 vs 0.325 ± 0.006).

Masking was negligible in coherent and null regimes but substantial in structured cases, with masked fractions of 0.278 ± 0.020 (opposing) and 0.281 ± 0.025 (phase), indicating that approximately 28% of genes were masked under structured regimes, defined by low-to-moderate pooled effect and high instability.

These results demonstrate that attenuation under pooling is not a consequence of noise or insufficient signal strength but arises specifically from structured disagreement between underlying responses. In such cases, pooled estimates provide no indication that biologically meaningful effects have been suppressed.

2.2. Instability is distinct from variance and reveals masked signals

To assess whether instability reveals signals not captured by pooled analysis, we compared pooled effect size and instability across 15,187 genes in the UV dataset (Fig. 3a–d). Using the thresholds $Q0.25(|\Delta\_pool|) = 0.119$ and $Q0.90(\text{instability}) = 1.146$, 283 genes (~1.9%) were identified within a masked region characterized by a low pooled effect but high instability. Representative genes (e.g., RvY_14730-1, RvY_14133, and RvY_06540-1; Supplementary Table S2) showed strong opposing subgroup effects ($|\Delta| \approx 1.18–1.31$) that collapsed toward near-zero when pooled.

Logistic regression confirmed instability as a strong predictor of masking ($\beta = 3.16 \pm 0.13$, $z = 24.28$, $p < 2 \times 10^{-16}$), whereas variance showed a weaker negative association ($\beta = -0.65 \pm 0.23$, $z = -2.86$, $p = 0.004$), improving overall model fit (Δdeviance = 847.8; AIC = 1973.1). This finding supports a distinction between variance as a measure of dispersion and instability as a measure of structured divergence.

Grouping genes by instability revealed a progressive increase in directional discordance. Median divergence rose from 0.40 (IQR: 0.18–0.72) in the full set to 1.33 (1.19–1.55) and 1.55 (1.42–1.77) in the top 10% and 5% instability tiers, respectively (~3.3-fold increase). Masked genes consistently exhibited opposing subgroup effects, indicating structured cancellation rather than a weak signal. DES-specific patterns are shown in Supplementary Fig. S1.

2.3. Instability differs across structured and homogeneous conditions

Instability distributions varied across conditions (Fig. 4a), with median values of 0.43 (UV), 0.39 (DES), and 0.77 (GAM), compared with 0.21 in the homogeneous LT condition. Structured conditions showed broader distributions and extended high-instability tails, whereas LT remained tightly constrained at low values, consistent with a homogeneous response.

Consistent with this pattern, high-instability tail occupancy was enriched in structured conditions (Fig. 4b). The fraction of genes within the top 10% instability range reached 10% (UV), 9% (DES), and 20% (GAM), compared with 0.8% in LT. Similarly, the top 5% tail was enriched in UV (4.8%), DES (5.3%), and GAM (9.6%), but remained minimal in LT (0.3%).

Decomposition of instability into heterogeneity gap and directional discordance revealed distinct condition-specific drivers (Fig. 4c). GAM exhibited the highest heterogeneity (0.67) and discordance (0.125), indicating strong temporal divergence. DES showed comparable heterogeneity (0.39) but negligible discordance, consistent with magnitude differences across subgroups. UV displayed intermediate heterogeneity (0.40) with modest discordance (0.026), reflecting partial subgroup conflict. In contrast, LT exhibited uniformly low heterogeneity (0.21) and near-zero discordance.

These structural differences translated into condition-specific masking behavior (Fig. 4d). The fraction of masked genes remained minimal in LT (0.0066%) but increased in structured conditions (UV 1.40%, DES 0.34%, GAM 3.37%; bootstrap means: 0.014 ± 0.002, 0.004 ± 0.001, 0.034 ± 0.003).

Together, these results show that instability is selectively elevated in the presence of structured internal variation and remains low in homogeneous conditions, indicating that it captures structured disagreement rather than overall signal magnitude.

2.4. Instability identifies genes missed by pooled analysis

Across all genes (n = 15,187), instability-based prioritization diverged strongly from pooled effect–based ranking (Fig. 5a). Rank–rank comparison revealed a distinct subset of genes that were highly prioritized by instability but ranked poorly by pooled log2 fold change, with representative genes (RvY_14730-1, RvY_18684-1, RvY_14091-1) shifting by more than 11,000 rank positions. This displacement indicates systematic masking of structured responses under aggregation.

Consistent with this displacement, instability- and pooled-based selections showed minimal overlap (Fig. 5b). At N = 100, no shared genes were observed, indicating distinct prioritization regimes. This separation reflects fundamentally different prioritization criteria rather than instability of ranking, with instability selectively capturing genes exhibiting structured heterogeneity, in contrast to pooled analysis, which favors genes with consistent average effects.

The separation between prioritization regimes remained stable across selection thresholds (Fig. 5c). Overlap remained negligible at N = 50 and N = 100, and increased only modestly at larger thresholds (Jaccard index $J(200) = 0.020$; $J(500) = 0.046$), corresponding to 8 and 44 shared genes, respectively, and remaining below 5% even under relaxed thresholds.

Representative examples illustrate the contrasting biological patterns captured by each approach (Fig. 5d). Instability-selected genes showed high instability (2.89–2.98) and strong effect divergence across groups (2.75–2.85), consistent with opposing or subgroup-dependent responses. In contrast, pooled-selected genes exhibited low instability (0.08–0.28) and consistent regulation across groups, reflecting stable average responses.

Functional characterization of instability-recovered masked genes further supports their biological relevance (Fig. 5e). Domain-based annotation (Pfam, InterProScan) revealed representation across multiple functional classes, including signaling/receptor proteins, metabolic enzymes, and nucleic acid-associated regulators, with complete annotation and classification provided in Supplementary Table S3. Notably, instability-based prioritization recovered neuroendocrine signaling peptides, including Corazonin (Crz), Pigment-Dispersing Hormone (PDH), and Crustacean Cardioactive Peptide (CCAP), which were not prominent under pooled analysis (Supplementary Table S3). These genes were identified within the masked set prior to functional interpretation, indicating that their prioritization was driven by structured instability rather than annotation completeness.

2.5. Instability reveals structured regulatory patterns in heterogeneous data

To test whether instability captures structured disagreement beyond variance, we performed controlled synthetic analysis across defined response regimes (Fig. 6). Median instability rose from 0.277 (coherent) to 1.815 (phase) and 3.410 (opposing), whereas variance increased from 0.034 to 0.506 and 2.292, and pooled effects decreased from 1.042 to 0.143 and 0.164.

Instability increased ~6.5-fold (phase) and ~12-fold (opposing), whereas variance increased ~67-fold, indicating nonproportional scaling. Variance reflects dispersion, whereas instability captures structured divergence across groups. Importantly, the nonproportional and scenario-dependent scaling indicates that instability is not a monotonic transformation of variance.

Together, these results show that instability quantifies internal disagreement rather than variance magnitude, providing a distinct signal for heterogeneous regulatory regimes.

3. Discussion

In this study, we demonstrate that in RNA-seq meta-analyses heterogeneity is an intrinsic property of dataset structure rather than a consequence of treatment–control design, and must be explicitly considered when interpreting pooled differential expression. When internal responses differ across subgroups or time, aggregation can suppress structured regulatory signals, collapsing opposing effects toward near-zero estimates and creating a systematic failure mode of pooled analysis[5-8].

Instability provides a quantitative measure of this internal disagreement prior to aggregation. Across synthetic and real datasets, instability increased selectively under opposing and phase-structured regimes while remaining low in coherent and null conditions. In real data, instability identified genes with subgroup-dependent or temporally structured responses that were not captured by pooled summaries. Unlike subgroup-specific analyses, which recover individual effects but do not quantify their compatibility, instability provides a unified measure of whether responses are coherent or conflicting across groups .

A defining property of instability is its condition-specific activation. It is elevated only when structured heterogeneity is present (UV, DES, GAM) and remains minimal in homogeneous conditions (LT). This behavior differentiates instability from variance, which reflects dispersion but cannot differentiate stochastic variability from structured disagreement17-20. Decomposition analysis further shows that instability arises from distinct structural drivers, including directional conflict, magnitude divergence, and temporal variation, supporting its interpretation as a structure-sensitive measure rather than a proxy for variance. Temporal regimes such as GAM may partially couple instability with response amplitude variation, indicating that some forms of temporal divergence can contribute variance-associated structure alongside directional disagreement (Supplementary Fig. S2).

A direct consequence of this behavior is divergence in gene prioritization. Instability-based ranking identifies genes largely distinct from those selected by pooled effect magnitude, reflecting fundamentally different prioritization criteria rather than instability of ranking. While pooled analysis preferentially selects genes with consistent directionality, instability highlights genes characterized by structured, subgroup-dependent, or temporally heterogeneous responses. As a result, genes that appear weak under pooled analysis may represent structured regulation that becomes detectable only through instability.

In conventional analysis workflows, such genes often require extensive post hoc exploration, including subgroup-specific contrasts or time-resolved analyses, to resolve underlying structure. Instability provides a direct measure of internal disagreement, enabling their identification without prior stratification or iterative contrast testing, thereby reducing reliance on ad hoc post hoc analysis.

The functional composition of the masked gene set further supports its biological relevance. Domain-based annotation of all masked genes (Supplementary Table S3) revealed multiple regulatory and signaling-associated features, including kinase/phosphatase modules, GPCR-related domains, and nucleic acid–associated components, consistent with stress-response pathways. Notably, instability-based prioritization recovered neuroendocrine signaling peptides, including Corazonin (Crz), Pigment-Dispersing Hormone (PDH), and Crustacean Cardioactive Peptide (CCAP), which were not prominent under pooled analysis. These genes were identified within the masked set prior to functional interpretation, indicating that their prioritization was driven by structured instability rather than annotation completeness. Given that such signaling molecules are often expressed in spatially restricted cell populations or exhibit temporally dynamic expression, their signals are susceptible to attenuation under pooled transcriptomic analysis. Their recovery is therefore consistent with structured regulatory heterogeneity that is not captured by aggregated summaries.

Instability quantifies structured disagreement of effect across defined groups, enabling the identification of genes whose regulatory behavior cannot be captured by a single pooled estimate. Rather than replacing differential expression analysis, it complements it by identifying when pooled summaries are incomplete or misleading. In heterogeneous datasets, pooled estimates can obscure structured regulatory patterns, leading to incorrect biological interpretation rather than reduced sensitivity. While subgroup-specific analyses can recover individual effects, they require prior knowledge of relevant structure and do not provide a unified measure of compatibility across groups. Instability addresses this gap by quantifying whether a single effect estimate is meaningful.

A limitation of the current framework is its reliance on predefined subgroup or temporal groups. In settings where structure is latent, additional approaches may be required to identify relevant groups before instability can be applied, as the method does not infer the source of heterogeneity but instead quantifies its presence and structure[13-16]. Incorrect or biologically irrelevant grouping can distort instability estimates and reduce their interpretability[13-16].

The applicability of instability extends to analyses where structured heterogeneity is encoded in experimental metadata, including subgroup, temporal, or condition-specific groups. Its utility is therefore highest in datasets where biologically meaningful structure is present, rather than in homogeneous systems.

Heterogeneous structure is common across biological systems, including stress responses, development, and mixed cell populations[21-24]. By quantifying internal disagreement directly, instability provides a principled way to distinguish absence of regulation from cancellation of conflicting effects, extending the interpretation of differential expression beyond pooled summaries.

4. Methods

Data sources and preprocessing

RNA-seq datasets were derived from stress-response experiments in the tardigrade *Ramazzottius varieornatus*, a model organism known for extreme tolerance to environmental stress. Publicly available SRA datasets included ultraviolet exposure (UV), desiccation (DES), gamma radiation (GAM), and low-temperature conditions (LT). Each dataset contained structured internal partitions defined either by biological subgroups (e.g., UV short vs long exposure; DES fast vs slow dehydration) or by temporal progression (GAM timepoints). LT was used as a homogeneous reference condition. Additional stress conditions, including high temperature (HT) and osmotic stress (OSM), are present in the dataset but are not central to the analyses used to demonstrate the instability framework, which focuses on conditions with well-defined internal structure.

Metadata were standardized using a custom preprocessing pipeline to harmonize sample identifiers, condition labels, treatment status, and partition structure (Supplementary Table S1)[3, 4]. The resulting curated metadata table was used consistently for all pooled and structured analyses.

Definition of homogeneous and heterogeneous conditions

A condition was defined as homogeneous when gene expression responses were consistent in both direction and magnitude across its internal samples (e.g., timepoints or replicates), such that pooled summaries closely approximated subgroup-aware analyses.

A condition was defined as heterogeneous when it contained structured internal variation, including subgroup-specific or time-dependent responses, leading to divergence or partial conflict across samples. In such cases, pooled summaries may obscure, attenuate, or distort the underlying signal.

Importantly, this definition of heterogeneity is independent of the presence of treatment and control groups. Each condition was analyzed relative to its own baseline control, corresponding to nonstressed samples (e.g., 0h timepoints or untreated controls within the same experiment). Although these controls represent a common biological baseline, they are defined independently within each dataset. The internal structure of each condition was then evaluated based on the consistency of gene expression responses across its samples. Conditions with consistent direction and magnitude were considered homogeneous, whereas those showing subgroup-specific or time-dependent divergence were classified as heterogeneous.

RNA-seq quantification and expression matrix construction

Transcript-level quantification was performed using Salmon. Gene-level expression estimates were obtained using tximport with a transcript-to-gene mapping (tx2gene.tsv)[25], followed by DESeq2 variance-stabilizing transformation (VST)[1] using default parameters (blind = FALSE). Principal component analysis (PCA) was performed on the VST-transformed matrix for exploratory analysis and visualization using the prcomp function in R.

All downstream analyses were performed on this gene-level VST matrix. Hereafter, aggregated expression features are referred to as genes.

Pooled and structured effect estimation

Pooled differential expression was implemented using a linear modeling framework (limma), in which all samples were analyzed jointly under a single treatment–control contrast without accounting for internal subgroup or temporal structure. Structured effects were estimated using the same framework, but treatment–control contrasts were computed separately within predefined partitions, including biological subgroups or timepoints.

For each gene *i*, differential expression was estimated as $\log_2$ fold change ($\log_2$FC). Let $\Delta_{i,k}$ denote the structured log_2 fold change for gene *i* in partition *k*, where *K* is the total number of partitions.

The pooled effect, $\Delta_{pool,i}$, was defined as the treatment–control effect estimated from the pooled model across all samples. This estimate reflects the aggregate treatment–control contrast across all samples, without preserving partition-specific structure.

For descriptive comparison across partitions, the mean structured effect for gene i was defined as:

$$\overline{\Delta}_i = \frac{1}{K}\sum_{k=1}^{K} \Delta_{i,k}$$

where *K* denotes the number of partitions.

Structured effects $\Delta_{i,k}$ were used to quantify internal divergence across partitions, whereas $\Delta_{pool,i}$ was used for masked-gene classification and pooled-ranking analyses.

Instability score

Instability was defined to quantify internal disagreement across structured partitions by combining effect divergence and directional inconsistency.

Effect divergence was defined as:

- for subgroup-based conditions:

$$D_i = | \Delta_{i,1} - \Delta_{i,2} |$$

- for temporal conditions:

$$D_i = \max_t (\Delta_{i,t}) - \min_t (\Delta_{i,t})$$

Directional discordance was defined as sign disagreement across partitions. For subgroup-based conditions:

$$C_i = \begin{cases} 1, & if \ sign(\Delta_{i,1}) \neq sign(\Delta_{i,2}) \\ 0, & otherwise \end{cases}$$

For temporal data, discordance was defined as the fraction of timepoints whose sign differed from the mean effect direction.

The instability score was then defined as:

$$I_i = D_i \cdot (1 + C_i)$$

This formulation captures magnitude divergence and directional inconsistency, which represent the core components of structured disagreement across partitions. This formulation increases with both magnitude divergence and directional conflict.

Masked gene definition

Genes were classified as masked by pooling if they exhibited weak pooled effect but strong internal disagreement. Let $Q_{0.25}$ denote the 25th percentile of $| \Delta_{pool} |$, and $Q_{0.90}$ the 90th percentile of instability. A gene *i* was defined as masked if:

$$| \Delta_{pool,i} | \leq Q_{0.25} \ \ and \ \ I_i \geq Q_{0.90}$$

The masked fraction within each condition was computed as the proportion of genes satisfying these criteria.

Thresholds were defined using fixed quantile cutoffs to identify regions of low pooled effect and high instability. Robustness of instability-based prioritization was assessed across a range of selection thresholds (Top-N) (Fig. 5B).

Variance proxy and regression analysis

To assess whether instability reflects dispersion alone, a variance proxy was computed from structured effect estimates as the variance of $\Delta_{i,k}$ across partitions[17, 18].

The relationship between masking and instability was evaluated using logistic regression[26]:

$$logit\left(P(M_i = 1)\right) = \beta_0 + \beta_1 I_i + \beta_2 V_i$$

where M_i indicates masked genes, I_i is instability, and V_i is the variance proxy.

Cross-condition analysis

Instability distributions were compared across UV, DES, GAM, and LT conditions. Enrichment of high-instability genes was quantified using empirical thresholds, including the fractions of genes exceeding the top 10% and top 5% instability levels.

Instability was further decomposed into its components (effect divergence and directional discordance) to characterize condition-specific contributions.

Gene prioritization and overlap analysis

Genes were ranked independently according to pooled effect magnitude and instability:

$$R_i^{pool} = rank(\ |\Delta_{pool,i}|\ ),\ R_i^{inst} = rank\left(I_i\right)$$

Top-N gene sets were defined for each ranking scheme:

$$S_N^{pool} = \{i{:}R_i^{pool} \le N\},\ S_N^{inst} = \{i{:}R_i^{inst} \le N\}$$

Instability-only and pooled-only sets were defined as set differences:

$$S_N^{inst-only} = S_N^{inst} \backslash S_N^{pool},\quad S_N^{pool-only} = S_N^{pool} \backslash S_N^{inst}$$

Overlap between sets was quantified using the Jaccard index:

$$J(N) = \frac{|\ S_N^{pool} \cap S_N^{inst}\ |}{|\ S_N^{pool} \cup S_N^{inst}\ |}$$

Analyses were performed for $N = 50, 100, 200,\ and\ 500$.

Representative genes were selected from instability-only and pooled-only sets based on objective criteria, including (i) rank displacement defined as $| R_i^{pool} - R_i^{inst} |$, (ii) magnitude of instability score, and (iii) consistency of structured effect patterns, defined as absence of stochastic fluctuations across partitions relative to the dominant trend. Selection was restricted to genes within the Top-N ranked sets used for overlap analysis, ensuring consistency with the global prioritization framework. No manual curation based on gene identity or prior functional knowledge was applied.

Functional annotation was assigned post hoc by mapping selected gene identifiers to reference genome annotations and associated protein records, followed by integration of annotations from NCBI, eggNOG, InterProScan, and Pfam databases.

Synthetic benchmarking and variance disentanglement

Synthetic datasets were constructed to represent coherent, opposing, and phase-structured regulatory regimes. For each scenario, pooled effect, instability, and variance proxy were computed as described above.

Summary statistics were reported as mean ± standard deviation or median and interquartile range. To assess whether instability is reducible to variance, changes in instability and variance were compared across scenarios.

Robustness and uncertainty estimation

Threshold robustness was evaluated across ranges of pooled-effect and instability quantiles. Stability of condition separation and gene prioritization was assessed across these parameter settings.

Uncertainty in masked fractions and related statistics was estimated using bootstrap resampling, and results were summarized as mean ± standard deviation.

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Competing Interests

The authors declare no competing interests.

Code and Data Availability

All analyses were performed using custom, fully documented R scripts. The complete analytical pipeline, including preprocessing, effect estimation, instability calculation, supplementary analyses, and figure generation, is maintained in a version-controlled GitHub repository (https://github.com/sajadshahbaz/NM2_instability_signal). The repository is currently private during peer review and will be made publicly available upon acceptance of the manuscript. A permanent archived Zenodo release and DOI will be generated at the time of public release. Processed datasets, metadata tables, precomputed resources required for figure generation, and intermediate outputs used to reproduce the manuscript figures are maintained within the repository. Raw RNA-seq datasets are publicly available through the Sequence Read Archive (SRA) under the BioProject accessions listed in Supplementary Table S1, comprising 111 samples across six independent studies. Metadata used for condition definition, partitioning, and downstream analyses are provided in Supplementary Table S1.

All results reported in this study can be reproduced using the supplied scripts, processed inputs, and default parameters. Computational environment specifications, software dependencies, and reproducibility instructions are documented within the repository. Repository access can be provided to editors and reviewers upon request during the peer-review process.

Author Contributions

S.S.: Conceptualization; Methodology; Data curation; Formal analysis; Investigation; Validation; Visualization; Writing – original draft, Writing – review & editing. **P.M.:** Funding acquisition; Project administration; Resources; Supervision; Writing – review & editing. **J.L.:** Investigation; Validation; Figure preparation; Writing – review & editing. **A.U.:** Scientific guidance; Writing – review & editing. **T.Z.:** Scientific guidance; Writing – review & editing.

12. Use of Generative AI Tools

Generative language models were used to assist with language editing, code formatting, and structural refinement during manuscript preparation (including Rubriq, American Journal Experts proofreading services). These tools were not used for algorithm design, parameter selection, data analysis, or interpretation of results. All methodological concepts, analytical decisions, and scientific conclusions were developed independently by the authors, who take full responsibility for the manuscript.

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